

TrES-3b

R-0469

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_200705 · created 2026-07-20T00:59:42+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459035.7829 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.064 +/- 0.053
Transit depth (Rp/Rs)^2	0.4 +/- 0.67 [%]
Orbital Inclination (inc)	82.48 +/- 2.85
Airmass coefficient 1 (a1)	0.886 +/- 0.011
Airmass coefficient 2 (a2)	0.112 +/- 0.072
Scatter in the residuals of the lightcurve fit is	1.25 %
Best Comparison Star	#3 - [483, 466]
Optimal Aperture	3.7
Optimal Annulus	10.92
Transit Duration (day)	0.024 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.064 ± 0.053 vs 0.16309 (deviation 1.75σ)
Duration fit vs published	0.024 d vs 0.0567 d (deviation 2.04σ)
Mid-time O–C	2.2 min
Depth SNR	1.1
Residual scatter	1.25 %

Light curve

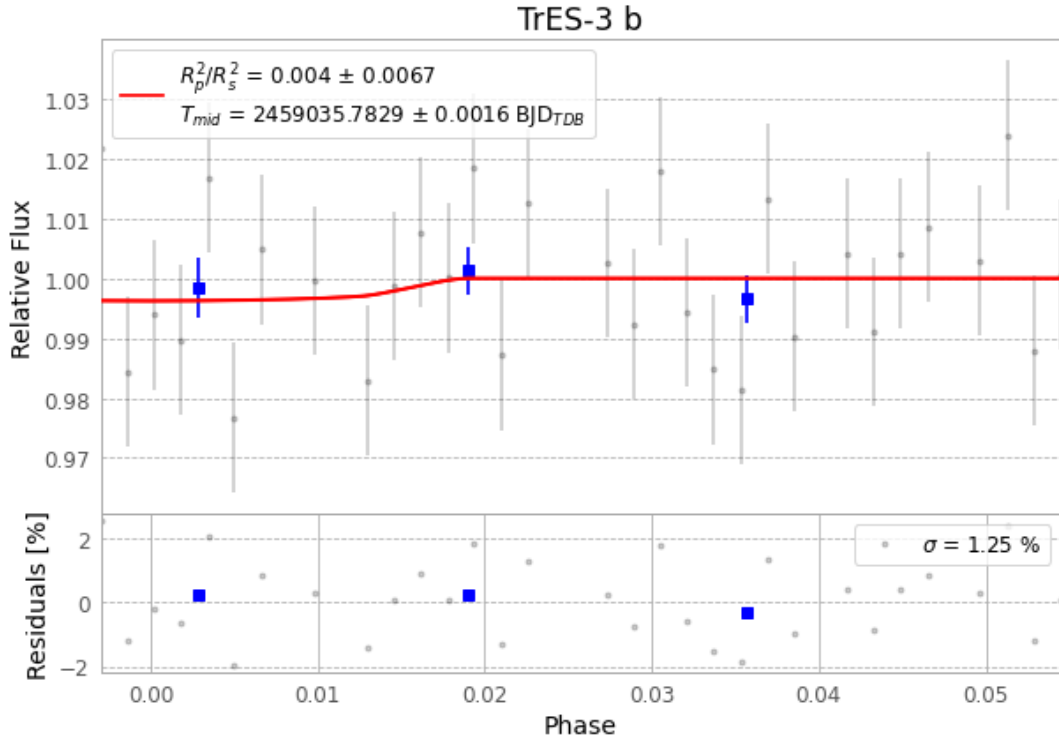


Figure 1 — FinalLightCurve_TrES-3 b_2020-07-04.png (SHA-256 d82eb2fdd4f947c1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [242, 249], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b6d619ab542e5133...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

32 FITS frames (21 MB), TRES-3200705063612.FITS → TRES-3200705082416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-200705.log (1d8aa8653763...), FinalLightCurve_TrES-3 b_2020-07-04.png (d82eb2fdd4f9...), FinalLightCurve_TrES-3 b_2020-07-04.csv (78bd65a89337...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 00:59Z.